Three-Router Triangle Network

Packet Tracer practical: DHCP + OSPF (Area 0)

Exam strategy: Build and test in layers. First interfaces and direct links, then OSPF, then DHCP, then PC-to-PC tests. Do not configure everything blindly at once.

1 1. Target topology

Three routers form a triangle. Each router connects to its own LAN through a switch, and each LAN receives addresses automatically through that LAN's router. OSPF advertises all LAN and transit networks so traffic can take an alternate path if one router-to-router link fails.

LAN 1	LAN 2	LAN 3
PCs – Switch – R1	PCs – Switch – R2	PCs – Switch – R3

```
                R1
               /  \\
              /    \\
          R2 ------ R3

R1 LAN: 192.168.10.0/26
R2 LAN: 192.168.10.64/26
R3 LAN: 192.168.10.128/26
All three router-to-router links use separate /30 networks.
```

Devices: 3 routers (e.g. 2911, with three GigabitEthernet ports), 3 switches, and at least 1 PC per LAN. Add more PCs if the task asks. Use copper straight-through from PC to switch and switch to router. For router-to-router Ethernet links, Packet Tracer's automatic connection tool is convenient; otherwise use the cable type your lab requires. Interface names and port numbers can differ by router model—record the actual ports you connect.

2 2. IP addressing plan

Use this table as the source of truth. Subnet mask for each LAN is `255.255.255.192` (/26); transit mask is `255.255.255.252` (/30).

Link	Device/interface	IP address	Mask	Purpose
LAN 1	R1 G0/0	192.168.10.1	255.255.255.192	PC gateway
LAN 2	R2 G0/0	192.168.10.65	255.255.255.192	PC gateway
LAN 3	R3 G0/0	192.168.10.129	255.255.255.192	PC gateway
R1–R2	R1 G0/1	10.0.0.1	255.255.255.252	Transit
	R2 G0/1	10.0.0.2	255.255.255.252	Same link
R2–R3	R2 G0/2	10.0.0.5	255.255.255.252	Transit
	R3 G0/1	10.0.0.6	255.255.255.252	Same link
R1–R3	R1 G0/2	10.0.0.9	255.255.255.252	Transit
	R3 G0/2	10.0.0.10	255.255.255.252	Same link

Port mapping is an example. If you connect a cable to a different port, use that actual interface name in the configuration. Both ends of a link must be in the same subnet, e.g. R1–R2 must be 10.0.0.1/30 and 10.0.0.2/30.

3 3. Build and bring up interfaces first

1. Place devices and rename them R1, R2, R3, SW1, SW2, SW3, PC1, PC2, PC3.
2. Connect each PC to its switch, each switch to its router's LAN interface, and connect R1–R2, R2–R3, and R1–R3.
3. On each router, enter the CLI and configure the interfaces below. If the interface names differ, substitute your real interface names.

R1 interface configuration

```
enable
configure terminal
hostname R1
interface gigabitEthernet0/0
 ip address 192.168.10.1 255.255.255.192
 no shutdown
exit
```

```
interface gigabitEthernet0/1
 ip address 10.0.0.1 255.255.255.252
 no shutdown
exit
interface gigabitEthernet0/2
 ip address 10.0.0.9 255.255.255.252
 no shutdown
exit
end
write memory
```

R2 interface configuration

```
enable
configure terminal
hostname R2
interface gigabitEthernet0/0
 ip address 192.168.10.65 255.255.255.192
 no shutdown
exit
interface gigabitEthernet0/1
 ip address 10.0.0.2 255.255.255.252
 no shutdown
exit
interface gigabitEthernet0/2
 ip address 10.0.0.5 255.255.255.252
 no shutdown
exit
end
write memory
```

R3 interface configuration

```
enable
configure terminal
hostname R3
interface gigabitEthernet0/0
 ip address 192.168.10.129 255.255.255.192
 no shutdown
exit
interface gigabitEthernet0/1
 ip address 10.0.0.6 255.255.255.252
 no shutdown
exit
interface gigabitEthernet0/2
 ip address 10.0.0.10 255.255.255.252
 no shutdown
exit
end
write memory
```

Checkpoint before OSPF: On every router run `show ip interface brief`. Connected ports should show `up/up`. If a port is down, check the cable and port first, then confirm `no shutdown`. If it is `administratively down`, `no shutdown` is missing.

4 4. Configure OSPF on all routers

Use OSPF process ID 1 and area 0 on all routers. The wildcard mask is the inverse of the subnet mask: /26 uses `0.0.0.63`; /30 uses `0.0.0.3`.

R1 OSPF

```
configure terminal
router ospf 1
 router-id 1.1.1.1
```

```
 network 192.168.10.0 0.0.0.63 area 0
 network 10.0.0.0 0.0.0.3 area 0
 network 10.0.0.8 0.0.0.3 area 0
 passive-interface gigabitEthernet0/0
end
write memory
```

R2 OSPF

```
configure terminal
router ospf 1
 router-id 2.2.2.2
 network 192.168.10.64 0.0.0.63 area 0
 network 10.0.0.0 0.0.0.3 area 0
 network 10.0.0.4 0.0.0.3 area 0
 passive-interface gigabitEthernet0/0
end
write memory
```

R3 OSPF

```
configure terminal
router ospf 1
 router-id 3.3.3.3
 network 192.168.10.128 0.0.0.63 area 0
 network 10.0.0.4 0.0.0.3 area 0
 network 10.0.0.8 0.0.0.3 area 0
 passive-interface gigabitEthernet0/0
end
write memory
```

The `passive-interface` command advertises the LAN subnet in OSPF but prevents OSPF hello packets from being sent toward PCs. Do not make the router-to-router interfaces passive.

Verify OSPF before DHCP

```
show ip ospf neighbor
show ip route
show ip protocols
```

Each router should discover its two directly connected router neighbors in a fully working triangle. In `show ip route`, OSPF-learned routes are marked with `O`. If neighbors are missing, check interface status, IP/mask on both ends, matching OSPF area, and correct `network` statements.

5 5. Configure DHCP on each router

DHCP is configured locally on each router for its own LAN. No DHCP relay / `ip helper-address` is needed because the DHCP server and clients share the same LAN.

R1 DHCP pool

```
configure terminal
ip dhcp excluded-address 192.168.10.1 192.168.10.10
ip dhcp pool LAN1
 network 192.168.10.0 255.255.255.192
 default-router 192.168.10.1
 dns-server 8.8.8.8
end
write memory
```

R2 DHCP pool

```
configure terminal
ip dhcp excluded-address 192.168.10.65 192.168.10.74
```

```
ip dhcp pool LAN2
 network 192.168.10.64 255.255.255.192
 default-router 192.168.10.65
 dns-server 8.8.8.8
end
write memory
```

R3 DHCP pool

```
configure terminal
ip dhcp excluded-address 192.168.10.129 192.168.10.138
ip dhcp pool LAN3
 network 192.168.10.128 255.255.255.192
 default-router 192.168.10.129
 dns-server 8.8.8.8
end
write memory
```

Excluded addresses are reserved so DHCP does not hand out the router gateway or the small reserved range. Each DHCP pool's network and mask must match that LAN's subnet.

6 6. Set PCs to DHCP and test

1. Click each PC → **Desktop** → **IP Configuration** → click **DHCP**.
2. PC1 should receive an address in `192.168.10.11-192.168.10.62`, gateway `192.168.10.1`.
3. PC2 should receive an address in `192.168.10.75-192.168.10.126`, gateway `192.168.10.65`.
4. PC3 should receive an address in `192.168.10.139-192.168.10.190`, gateway `192.168.10.129`.
5. On each PC, open Desktop → Command Prompt and run `ipconfig`. Then ping the local gateway and PCs in the other LANs.

```
PC> ipconfig
PC> ping 192.168.10.1       (from a PC in LAN 1)
PC> ping <another-PC-IP>    (test across LANs)
```

Recommended test order: (1) each PC can ping its gateway, (2) router-to-router links and OSPF neighbors work, (3) each router has OSPF routes to both remote LANs, (4) PC1 pings PC2 and PC3, and PC2 pings PC3.

7 7. Fast troubleshooting table

Symptom	Likely cause	What to check
Interface shows down/down	Cable, wrong port, or peer interface down	Physical connections; `show ip interface brief`; correct interface names
Interface shows administratively down	Interface is shut down	Enter interface mode and run `no shutdown`
No OSPF neighbors	Wrong IP/mask, wrong OSPF network statement/area, link down	Compare both ends of each transit /30; run `show ip ospf neighbor`
PC gets 169.254.x.x or no IP	DHCP pool typo, wrong LAN interface, DHCP not enabled on PC	Verify pool network/mask and gateway; click DHCP again; inspect `show ip dhcp binding`
PC can ping gateway but not remote LAN	OSPF missing or remote route absent	Run `show ip route`; look for `O` routes; inspect OSPF neighbors
One PC fails but others work	PC/cable/switch port or wrong static setting	Check that PC is set to DHCP and its IP/gateway match the correct LAN

Useful commands:

```
show ip interface brief
show ip ospf neighbor
show ip route
show ip protocols
show ip dhcp pool
show ip dhcp binding
show running-config
```

```
ping <destination-IP>
```

8 8. Exam checklist and memory hooks

- **Triangle = 3 transit links:** R1–R2, R2–R3, R1–R3. Each link needs its own subnet.
- **/30 transit subnet:** mask `255.255.255.252`, wildcard `0.0.0.3`; exactly two usable host addresses.
- **/26 LAN subnet:** mask `255.255.255.192`, wildcard `0.0.0.63`; 62 usable host addresses.
- **Every LAN needs a unique subnet** and the router LAN IP is the DHCP default gateway.
- **OSPF process ID** can be locally chosen; here it is 1. All neighboring interfaces must participate in the same area (area 0 here).
- **DHCP pool pattern:** exclude gateway/reserved IPs → create pool → set network/mask → set default-router → optional DNS.
- Save configs with `write memory` (or `copy running-config startup-config`).

9 9. If the question changes slightly

- If the task specifies different IP ranges, keep the same method but replace every IP, subnet mask, wildcard mask, DHCP network, and gateway consistently.
- If the router has only two GigabitEthernet ports, use a suitable router model/module or use serial interfaces if permitted. Serial links need serial interface commands and the DCE end may need a clock rate; do not paste Ethernet interface names onto serial ports.
- If the question asks for a single DHCP server on a different LAN, clients on other LANs will need DHCP relay configured on their LAN interfaces using `ip helper-address <DHCP-server-IP>`. That is different from this guide's design, where each router serves its own LAN.

Final sanity check: interfaces up/up → OSPF neighbors/routes → DHCP leases → cross-LAN pings.
Good luck. Configure one layer at a time; verify before moving on.